

Lab 1: Use-Case Flow Specification

Use Case: UC-06 Dispatch Budget Breach Alert

Primary Actor: FinOps Analyst (recipient)

Secondary Actors: Cloud Administrator, Notification Service (email/Slack)

Preconditions

- A budget threshold has been configured for the project (UC-05).
- Billing data for the current cycle has been ingested and categorized (UC-01, UC-02).
- The FinOps Analyst has a valid, verified email address and/or Slack handle on file.

Postconditions

- Success: The responsible FinOps Analyst has received a breach alert containing the project name, threshold, and overage amount, and the alert is logged with a timestamp.
- Failure: The breach is logged as detected but undelivered, and the case is escalated to the Cloud Administrator / manager for manual follow-up.

Main Success Scenario

1. The system recalculates a project's month-to-date spend after each billing ingestion cycle.
2. The system compares month-to-date spend against the project's configured budget threshold.
3. The system detects that spend has exceeded the threshold and marks the project as “breached.”
4. The system composes an alert containing the project name, threshold, current spend, and overage percentage.
5. The system sends the alert to the responsible FinOps Analyst via the configured channel (email and/or Slack).
6. The notification service confirms successful delivery.
7. The system logs the alert (project, timestamp, channel, delivery status) and marks the breach as “notified.”
8. Use case ends successfully.

Alternate Flow

6a. Notification Delivery Failure

- 6a1. The notification service reports that delivery failed (e.g., invalid address, channel outage).
- 6a2. The system retries delivery once after a short delay.
- 6a3. If the retry also fails, the system extends this use case with «extend» “Escalate Breach to Manager,” notifying the Cloud Administrator so a human can follow up directly.
- 6a4. The system logs the failed delivery attempts and the escalation, then ends the use case.